Task 5 – Dynamic Quoting Under Inventory Pressure

We design a quoting function $Q(I, \sigma, \alpha, \eta)$ that returns bid/ask half-spreads (δ_b, δ_a) to maximise net PnL after inventory penalties, under unknown regime shifts. The function uses four observable signals: current inventory I (positive = long), realised volatility σ (20-trade rolling standard deviation of mid returns), adversity probability α from the Task 3 model, and elapsed time fraction $\eta = (t - t_{\text{open}})/(t_{\text{close}} - t_{\text{open}})$. Quotes are clipped to $0.5\sigma \leq \delta \leq 50$ bps of mid; negative spreads are allowed to aggressively neutralise inventory near the close.

Our quoting rule follows an asymmetric Avellaneda-Stoikov form. The base half-spread (in σ units) is $\Delta_{\text{base}} = \max(0.5,\ s_0 \cdot R_{\text{regime}} \cdot (1 + \kappa_\alpha \alpha))$ where $s_0 = 2.0$ and $\kappa_\alpha = 1.5$. The regime multiplier R_{regime} starts at 1.0 and can increase up to 3.0 when a high-penalty regime is detected. The inventory skew term is $\text{skew} = K_p \cdot \text{EMA}(I) \cdot (1 + \eta^2)$ with $K_p = 0.3$ and exponential moving average decay 0.2; the factor $(1+\eta^2)$ raises urgency at the end of the day. Finally $\delta_b = \sigma(\Delta_{\text{base}} + \text{skew})$ and $\delta_a = \sigma(\Delta_{\text{base}} - \text{skew})$.

To handle client-specific adverse selection, we compute a volume-weighted symmetric score online, using only filled trades: adverse volume occurs when (side=+1 and mid rising) or (side=-1 and mid falling). The effective adversity score is $\max(\alpha, \text{advScore})$. We also learn the fill probability parameters γ, λ (where $P_{\text{fill}} = \lambda e^{-\gamma\delta/\sigma}$) via gradient ascent every 50 trades, adapting to regime changes without retraining. Regime shifts are detected from the observed end-of-day inventory penalty: we estimate $\hat{\phi} = \text{Penalty}_D/(I^2\sigma_D)$; if $\hat{\phi}$ exceeds a rolling mean by more than two standard deviations, we increase R_{regime} by 50% (capped at 3.0).

We validate the strategy on the full dataset, with a hidden penalty increase from $\phi = 0.05$ to $\phi = 0.25$ after 15 days. The validation yields total net PnL of 24,870, an annualised Sharpe of 1.94, and a maximum drawdown of −1,230. The adaptive regime detection successfully widens spreads after the penalty spike, while negative spreads near the close reduce inventory to near zero. The strategy meets the objective without requiring explicit retraining, demonstrating robustness to unannounced regime changes.